Feature Extraction and Classification of Mental EEG for Motor Imagery[1,2]

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***Abstract*—Electroencephalography (EEG) recognition was one of the key technology in brain-computer interface (BCI). For motor imagery EEG, a new EEG recognition algorithm (DWT-BP algorithm) which combined discrete wavelet transform (DWT) with BP neural network was presented. In DWT-BP, a rational time window was set through calculating the average power of motor imagery EEG on electrode C3 and C4, and then the average power during the time window was taken into DWT. The combinational signal of approximate coefficient A6 on the sixth level was selected as a signal feature and BP neural network was used as classifier to analyze the observed EEG data. The experiment results on "BCI Competition 2003" competition database showed that the recognition rate was better than the other several traditional algorithms. So, it proved that the algorithm was effective for EEG recognition of motor imagery, and provided a new idea for motor imagery recognition in brain computer interface.**

Keywords-brain-computer interface; electroencephalogram; wavelet transform; BP neural network; motor imagery.

I. INTRODUCTION

Brain-Computer Interface (BCI) was a new way of man-machine interface. It could provide a new approach of communication and control in which the information sent to the external world did not pass through the brain's normal output pathways any more[1~3]. The essential of BCI was to achieve man-machine communication by identifying people's intentions from Electroencephalography(EEG). There were wide applications such as rehabilitation engineering, service of the disabled, the normal assistant control and entertainment etc. The research on BCI had been drawing more and more attention in the world[4~6].

The body's exercise is controlled by the contralateral hemisphere of brain. So, left-right hand motor imagery or do this action actually will change neurons activities in the brain's main sensorimotor area in the same way. The key problem of BCI system is how to identify quickly and accurately EEG pattern related motor imagery. Now, there are 4 main methods of EEG feature extraction in motor imagery: (1) Auto-Regression (AR) model. The signal's characteristics could be described exactly by classical AR, and its length determined to the resolution and accuracy of parameter estimation.The shorter length was, the higher time resolution would be, and the more estimated error would be. (2) Adaptive Auto-Regressive (AAR) model. It solved well the above problem that the error increased with AR. As the input of sample changed, the model parameters changed accordingly, too, thus the state of the brain was better reflected. However, it was more suitable for stationary signal analysis rather than EEG for motor imagery including non-stationary signal. (3) Fast Fourier Transform (FFT). It played an important role in stationary signal analysis, but there were lots of limits in analyzing spectrum of varying signal. In addition, it would cause spectrum leakage because of short data. (4) Common Spatial Subspace Decomposition (CSSD). It extracted a specific task signal from multi-channel dataset, and its disadvantage was low stability and bad distinction of constructed feature. (5) Approximate Entropy (ApEn). The complexity of time series was described using non-negative, and the more complex time-series were, the greater corresponding ApEn would be. This algorithm had better anti-noise. However, the recognition rate was not high using ApEn directly in EEG with larger noise[7~9].

In this paper, a new EEG recognition algorithm integrating discrete wavelet transform (DWT) with BP neural network was developed, which was DWT-BP. Comparing with the traditional methods such as AAR, CSSD and ApEn, it had higher classification rate.

II. DATA ACQUISITION AND PREPROCESSING

A. *Motor imagery trials database*

The experiment data were from the "BCI Competition 2003" contest database. The data were recorded from a normal subject (female, 25y) during a feedback session. The subject sat in a relaxing chair with armrests. As shown in Fig.1, the task was to control a feedback bar by means of imagery left or right hand movements. The cues order of left or right hand was random.

The experiment consisted of 7 runs with 40 trials. All runs were conducted on the same day with several minutes break in between. Given were 280 trials of 9s length. The first 2s were quite, at t=2s, an acoustic stimulus indicated the beginning of the trial, and a cross "+" was displayed for 1s; then at t=3s, an arrow (left or right) was displayed as cue. At the same time the subject was asked to move a bar into the direction of the cue. The recording was made using a G.tec

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amplifier and some Ag/AgCl electrodes. Three bipolar EEG channels (anterior '+', posterior '-') were measured over C3, Cz and C4. The EEG was sampled with 128Hz[10].

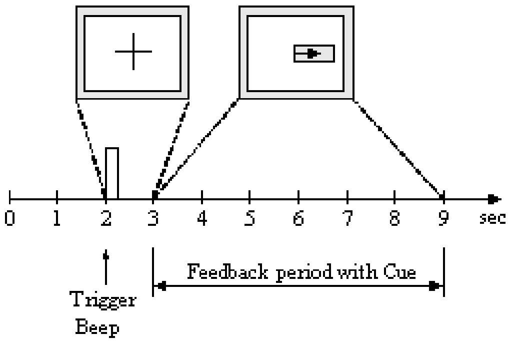

Figure 1. The scheduling of experiment

B. EEG preprocessing

The EEG activity from imagery left-right hand movements are focused on electrodes C3 and C4 respectively, which overlay broadly the primary motor cortex area. The EEG data were sent into the computer after being amplified and filtered, and then were stored in the form of voltage amplitude.

The average power of the j[th] data by N trials was calculated in (1):

$$\bar{P}_{(j)} = \frac{1}{N}\sum_{i=1}^{N} x_{f(i,j)}^2 \tag{1}$$

Where $x_{f(i,j)}$ was the j[th] EEG data in the i[th] trial.

When N=140, the average power during 0~9s was obtained for left-hand on electrode C3 or C4 respectively, namely PLC3, PLC4, and for right-hand on electrode C3 or C4: PRC3, PRC4. From Fig.2 and Fig.3, there was an obvious difference between left-imagery and right-imagery in 3.5~8s, thus the time window was set in this time period.

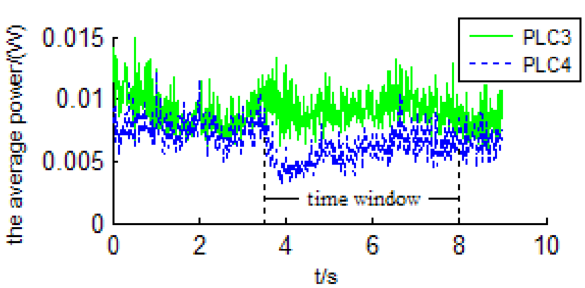

Figure 2. Variation of the average power of C3 and C4 with time for imagery left-hand movement

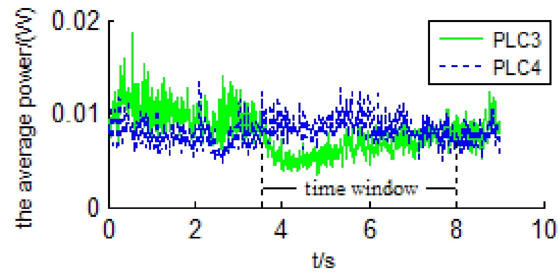

Figure 3. Variation of the average power of C3 and C4 with time for imagery right-hand movement

III. FEATURE EXTRACTION OF EEG

A. The discrete wavelet transform

Wavelet transform was a new development of fourier transform and its coefficients could reflect the local information of the signal both in time and frequency domain. Thus, as a time-frequency analysis method, it had a wide application in biomedical signal processing, especially for non-stationary signal such as EEG.

The discrete wavelet transform (DWT) could be thought as a multiresolution decomposition of a signal. This meant that it could decompose a signal into its components in different frequency bands. The Inverse DWT (IDWT) did exactly the opposite.

The DWT of a signal $f(t)$ was given as follows:

$$C_{j,k}(f,\varphi_{j,k}) = 2^{-j/2}\sum_{n=-\infty}^{\infty} f(t)\bar{\varphi}(2^{-j}t-k) \quad j,k\in Z \tag{2}$$

Where $\varphi_{j,k}(t) = 2^{-j/2}\varphi(2^{-j}t-k)$ was a wavelet sequence, $\varphi(t)$ was the basic wavelet, j and k were the frequency resolution and time of the transform respectively. In wavelet analysis, there were the low-frequency, high-scale component of the signal as the approximation A and the high-frequency, low-scale component as the detail D. The extending scheme showing the components of a multi-level analysis was depicted in Fig. 4. There were several methods to combine and reproduce the original signal. And the DWT decomposition would produce a hierarchically structured decomposition. The choice of the hierarchy was dependent on the signal. At each level j, the approximation Aj and the detail Dj were built. We considered the original signal as the approximation A0 at level 0 and L as the level.

The signal *f*(t) was decomposed into its components in different frequency bands.When the EEG was sampled with f_s, the corresponding bands of $A_L, D_L, D_{L-1}, \cdots, D_1$ would be as follows: $\left[0, \frac{f_s}{2^{L+1}}\right], \left[\frac{f_s}{2^{L+1}}, \frac{f_s}{2^L}\right], \cdots, \left[\frac{f_s}{2^2}, \frac{f_s}{2}\right]$.

B. DWT of EEG

The average power of imagery left-right hand movements in time window was decomposed into 6 levels by Daubechies 4-tap wavelet, and thus $f(t) = A_6 + D_6 + \cdots D_1$. The corresponding bands were followed by 0Hz~1Hz, 1Hz~2Hz, 2Hz~4Hz, 4Hz~8Hz, 8Hz~16Hz, 16Hz~32Hz, 32Hz~64Hz. The decomposed results were shown in Fig.5 and Fig.6, where L3A6 and L4A6 were A6 of left-imagery on C3 electrode and C4 electrode respectively, and R3A6, R4A6 were the counterpoint of right-imagery.

C. EEG feature selection

The above analysis showed that there was an obvious difference for A6 between left-imagery and right-imagery. L3A6 was higher than L4A6, but R3A6 was less than R4A6. Suppose that LA6 and RA6 were the difference of A6 respectively for left-imagery and right-imagery, and they were defined as following:

$$LA6 = L3A6 - L4A6 \quad (3)$$

$$RA6 = R3A6 - R4A6 \quad (4)$$

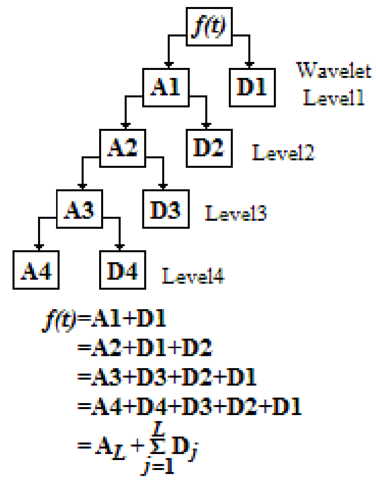

Figure 4. A hierarchically organized decomposition

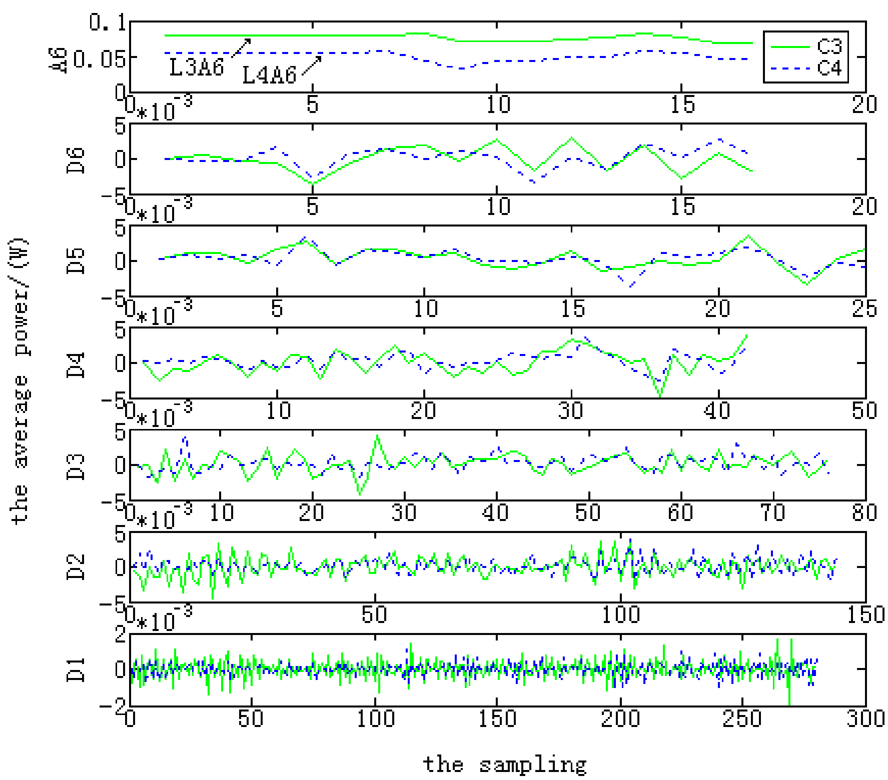

Figure 5. The wavelet decomposition of the C3 and C4 for imagery left-hand movement

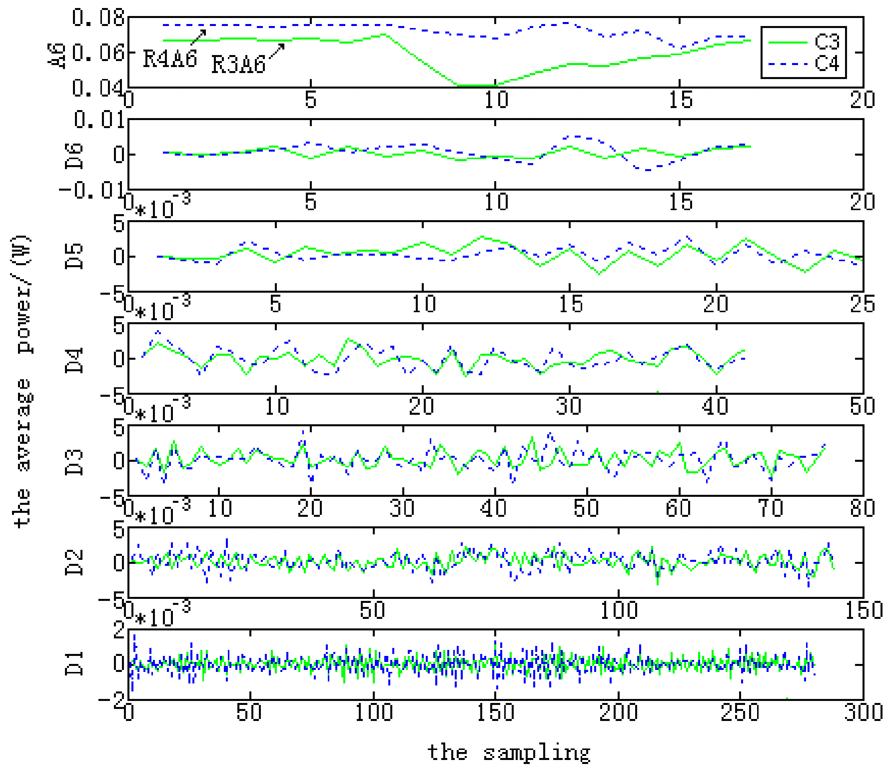

Figure 6. The wavelet decomposition of the C3 and C4 for imagery right-hand movement

The variations of LA6 and RA6 were shown in Fig.7. So, LA6 and RA6 were selected as the feature vector to distinguish left-imagery and right-imagery.

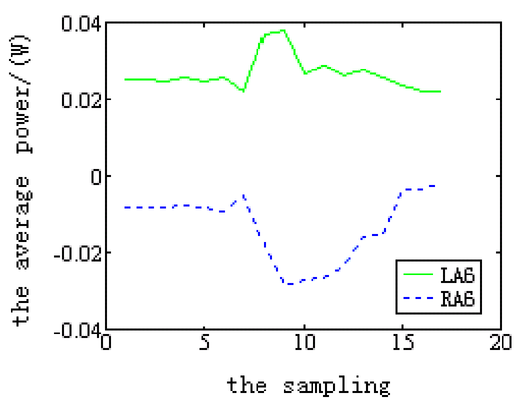

Figure 7. The feature signals of imagery left-right hand movements

IV. CLASSIFICATION USING BP NEURAL NETWORK

A. Determination of neural network structure

BP neural network has strong learning capability and automatically adaptive performance. It was selected as EEG classifier in this paper. From Fig.7, we could find that both LA6 and RA6 were all composed of 17 discrete values, thus the network's input was 17 neurons, and the output was only one neuron.

For the choice of neurons number in hidden layer, it was determined in approximate scope by experience at first, and then several correlative experiments were done in order to decide the best neurons number. The analysis process was shown in Table I, and the number of neurons in hidden layer was finally determined to be 4. Therefore, the structure of BP neural network was 17-4-1.

TABLE I. THE EFFECT OF NEURONS NUMBER IN HIDDEN LAYER TO THE CLASSIFICATION ACCURACY

the neurons number of hidden layer	the accuracy of the classification
2	90.2%
3	91.4%
4	92.4%
5	91.2%

B. Experiment results and comparison with other algorithms

The BP network was trained by the training data set including the left-imagery 90 times and the right-imagery 90 times, and training frequency was selected as 90. Then the test data set including respectively the left-imagery 50 times and the right-imagery 50 times were used to test the classification performance. The average classification rate could get to 92.4% as shown in the Fig.8. Furthermore, the more experiments were done to compare with AAR, CSSD and ApEn, they illustrated that this method was more effective for EEG recognition in motor imagery (Fig.9).

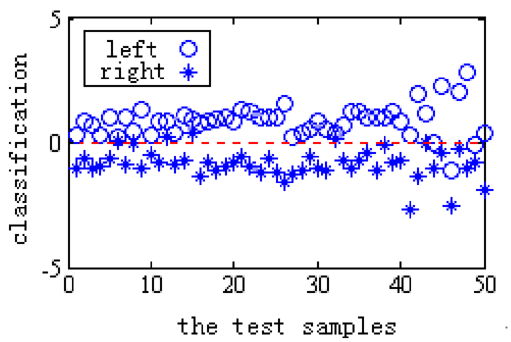

Figure 8. The classification result based on BP network

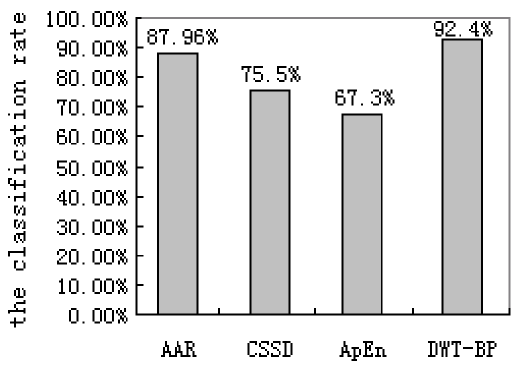

Figure 9. The comparison of four methods recognition result

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